SE LAB-06

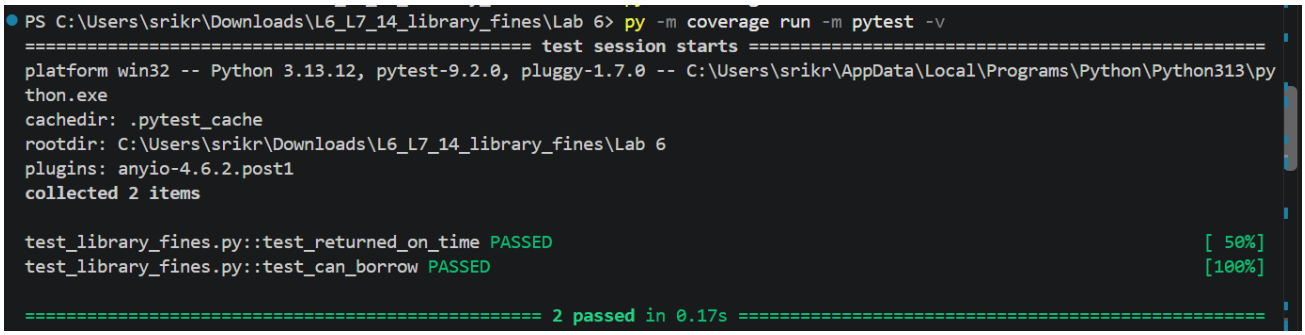
Github link :

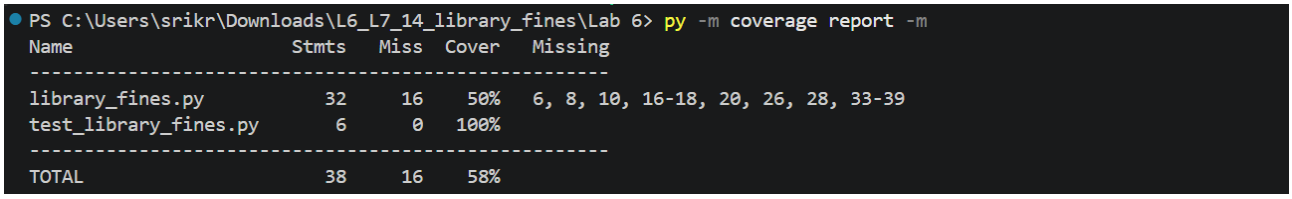
https://github.com/kruthivankadara-dev/SE-Lab-Submissions

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SRN : PES2UG24CS574

SEC : 5J





```
PS C:\Users\srikr\Downloads\L6_L7_14_library_fines\Lab 6> py -m coverage erase
PS C:\Users\srikr\Downloads\L6_L7_14_library_fines\Lab 6> py -m coverage run -m pytest -v CodeCoverageTest_PES2UG24CS574_
application.py
============================================= test session starts =============================================
platform win32 -- Python 3.13.12, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\srikr\AppData\Local\Programs\Python\Python313\py
thon.exe
cachedir: .pytest_cache
rootdir: C:\Users\srikr\Downloads\L6_L7_14_library_fines\Lab 6
plugins: anyio-4.6.2.post1
collected 21 items

CodeCoverageTest_PES2UG24CS574_application.py::test_unknown_member PASSED                                [  4%]
CodeCoverageTest_PES2UG24CS574_application.py::test_negative_days PASSED                                 [  9%]
CodeCoverageTest_PES2UG24CS574_application.py::test_student_on_time PASSED                               [ 14%]
CodeCoverageTest_PES2UG24CS574_application.py::test_student_overdue PASSED                               [ 19%]
CodeCoverageTest_PES2UG24CS574_application.py::test_faculty_overdue PASSED                               [ 23%]
CodeCoverageTest_PES2UG24CS574_application.py::test_guest_overdue PASSED                                 [ 28%]
CodeCoverageTest_PES2UG24CS574_application.py::test_fine_capped_at_200 PASSED                            [ 33%]
CodeCoverageTest_PES2UG24CS574_application.py::test_damaged_book_without_overdue PASSED                  [ 38%]
CodeCoverageTest_PES2UG24CS574_application.py::test_damaged_book_with_overdue PASSED                     [ 42%]
CodeCoverageTest_PES2UG24CS574_application.py::test_damaged_book_with_capped_fine PASSED                 [ 47%]
CodeCoverageTest_PES2UG24CS574_application.py::test_pending_fine_above_100 PASSED                        [ 52%]
CodeCoverageTest_PES2UG24CS574_application.py::test_pending_fine_exactly_100 PASSED                      [ 57%]
CodeCoverageTest_PES2UG24CS574_application.py::test_faculty_can_borrow PASSED                            [ 61%]
CodeCoverageTest_PES2UG24CS574_application.py::test_faculty_reached_book_limit PASSED                    [ 66%]
CodeCoverageTest_PES2UG24CS574_application.py::test_student_can_borrow PASSED                            [ 71%]
CodeCoverageTest_PES2UG24CS574_application.py::test_student_reached_book_limit PASSED                    [ 76%]
CodeCoverageTest_PES2UG24CS574_application.py::test_renewal_denied_due_to_reservation PASSED             [ 80%]
CodeCoverageTest_PES2UG24CS574_application.py::test_renewal_limit_reached PASSED                         [ 85%]
CodeCoverageTest_PES2UG24CS574_application.py::test_renewal_overdue PASSED                               [ 90%]
CodeCoverageTest_PES2UG24CS574_application.py::test_renewal_successful PASSED                            [ 95%]
```

```
PS C:\Users\srikr\Downloads\L6_L7_14_library_fines\Lab 6> py -m coverage run -m pytest -v CodeCoverageTest_PES2UG24CS574_
application.py
CodeCoverageTest_PES2UG24CS574_application.py::test_student_reached_book_limit PASSED                    [ 76%]
CodeCoverageTest_PES2UG24CS574_application.py::test_renewal_denied_due_to_reservation PASSED             [ 80%]
CodeCoverageTest_PES2UG24CS574_application.py::test_renewal_limit_reached PASSED                         [ 85%]
CodeCoverageTest_PES2UG24CS574_application.py::test_renewal_overdue PASSED                               [ 90%]
CodeCoverageTest_PES2UG24CS574_application.py::test_renewal_successful PASSED                            [ 95%]
CodeCoverageTest_PES2UG24CS574_application.py::test_reservation_takes_priority PASSED                    [100%]

============================================== 21 passed in 0.19s ==============================================
PS C:\Users\srikr\Downloads\L6_L7_14_library_fines\Lab 6> py -m coverage report -m library_fines.py
Name               Stmts   Miss  Cover   Missing
------------------------------------------------
library_fines.py      32      0   100%
------------------------------------------------
TOTAL                 32      0   100%
```

```
PS C:\Users\srikr\Downloads\L6_L7_14_library_fines\Lab 6> py -m coverage erase
PS C:\Users\srikr\Downloads\L6_L7_14_library_fines\Lab 6> py -m coverage run --branch -m pytest -v BranchCoverageTest_PES
2UG24CS574_application.py
============================================= test session starts =============================================
platform win32 -- Python 3.13.12, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\srikr\AppData\Local\Programs\Python\Python313\py
thon.exe
cachedir: .pytest_cache
rootdir: C:\Users\srikr\Downloads\L6_L7_14_library_fines\Lab 6
plugins: anyio-4.6.2.post1
collected 13 items

BranchCoverageTest_PES2UG24CS574_application.py::test_member_valid_and_invalid PASSED                    [  7%]
BranchCoverageTest_PES2UG24CS574_application.py::test_days_valid_and_invalid PASSED                      [ 15%]
BranchCoverageTest_PES2UG24CS574_application.py::test_faculty_and_non_faculty_loan_periods PASSED        [ 23%]
BranchCoverageTest_PES2UG24CS574_application.py::test_overdue_branch_and_fine_cap PASSED                 [ 30%]
BranchCoverageTest_PES2UG24CS574_application.py::test_fine_cap_boundary PASSED                           [ 38%]
BranchCoverageTest_PES2UG24CS574_application.py::test_damaged_and_undamaged_books PASSED                 [ 46%]
BranchCoverageTest_PES2UG24CS574_application.py::test_pending_fine_branch PASSED                         [ 53%]
BranchCoverageTest_PES2UG24CS574_application.py::test_faculty_borrowing_branches PASSED                  [ 61%]
BranchCoverageTest_PES2UG24CS574_application.py::test_non_faculty_borrowing_branches PASSED              [ 69%]
BranchCoverageTest_PES2UG24CS574_application.py::test_reservation_branch PASSED                          [ 76%]
BranchCoverageTest_PES2UG24CS574_application.py::test_renewal_limit_branch PASSED                        [ 84%]
BranchCoverageTest_PES2UG24CS574_application.py::test_overdue_branch PASSED                              [ 92%]
BranchCoverageTest_PES2UG24CS574_application.py::test_renewal_condition_priority PASSED                  [100%]

============================================== 13 passed in 0.16s ==============================================
```

```
PS C:\Users\srikr\Downloads\L6_L7_14_library_fines\Lab 6> py -m coverage report -m library_fines.py
Name              Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------------
library_fines.py      32      0     22      0   100%
-------------------------------------------------------------
TOTAL                 32      0     22      0   100%
PS C:\Users\srikr\Downloads\L6_L7_14_library_fines\Lab 6>
```

CodeCoverageTest_PES2UG24CS574_application.py :

```python
import pytest
from library_fines import late_fine, can_borrow, renewal_status
# ------------------------------------------------
# 1. TEST CASES FOR late_fine()
# ------------------------------------------------
def test_unknown_member():
    with pytest.raises(ValueError, match="unknown member"):
        late_fine("visitor", 20, False)
def test_negative_days():
    with pytest.raises(ValueError, match="invalid days"):
        late_fine("student", -1, False)
def test_student_on_time():
    assert late_fine("student", 14, False) == 0
def test_student_overdue():
    assert late_fine("student", 15, False) == 2
def test_faculty_overdue():
    assert late_fine("faculty", 31, False) == 1
def test_guest_overdue():
    assert late_fine("guest", 15, False) == 5
def test_fine_capped_at_200():
    assert late_fine("student", 200, False) == 200
```

```python
def test_damaged_book_without_overdue():
    assert late_fine("student", 10, True) == 150
def test_damaged_book_with_overdue():
    assert late_fine("student", 15, True) == 152
def test_damaged_book_with_capped_fine():
    assert late_fine("student", 200, True) == 350
# ------------------------------------------------
# 2. TEST CASES FOR can_borrow()
# ------------------------------------------------
def test_pending_fine_above_100():
    assert can_borrow(0, 101, "student") is False
def test_pending_fine_exactly_100():
    assert can_borrow(0, 100, "student") is True
def test_faculty_can_borrow():
    assert can_borrow(9, 0, "faculty") is True
def test_faculty_reached_book_limit():
    assert can_borrow(10, 0, "faculty") is False
def test_student_can_borrow():
    assert can_borrow(2, 0, "student") is True
def test_student_reached_book_limit():
    assert can_borrow(3, 0, "student") is False
# ------------------------------------------------
# 3. TEST CASES FOR renewal_status()
# ------------------------------------------------
def test_renewal_denied_due_to_reservation():
    assert renewal_status(0, True, 5) == "denied"
def test_renewal_limit_reached():
    assert renewal_status(2, False, 5) == "limit reached"
def test_renewal_overdue():
    assert renewal_status(1, False, -1) == "overdue"
def test_renewal_successful():
    assert renewal_status(0, False, 5) == "renewed"
def test_reservation_takes_priority():
    assert renewal_status(2, True, -1) == "denied"
```

BranchCoverageTest_PES2UG24CS574_application.py :

```python
import pytest
from library_fines import late_fine, can_borrow, renewal_status
# ------------------------------------------------
# 1. BRANCH TESTS FOR late_fine()
# ------------------------------------------------
def test_member_valid_and_invalid():
    with pytest.raises(ValueError, match="unknown member"):
        late_fine("visitor", 10, False)
```

```
    assert late_fine("student", 10, False) == 0
def test_days_valid_and_invalid():
    with pytest.raises(ValueError, match="invalid days"):
        late_fine("student", -1, False)

    assert late_fine("student", 14, False) == 0
def test_faculty_and_non_faculty_loan_periods():
    assert late_fine("faculty", 30, False) == 0
    assert late_fine("faculty", 31, False) == 1
    assert late_fine("student", 14, False) == 0
    assert late_fine("guest", 15, False) == 5
def test_overdue_branch_and_fine_cap():
    assert late_fine("student", 14, False) == 0
    assert late_fine("student", 15, False) == 2
    assert late_fine("student", 200, False) == 200
def test_fine_cap_boundary():
    # 100 overdue days * Rs. 2 = Rs. 200
    assert late_fine("student", 114, False) == 200

    # 101 overdue days * Rs. 2 exceeds Rs. 200
    assert late_fine("student", 115, False) == 200
def test_damaged_and_undamaged_books():
    assert late_fine("student", 10, False) == 0
    assert late_fine("student", 10, True) == 150
    assert late_fine("student", 15, True) == 152
# ------------------------------------------------
# 2. BRANCH TESTS FOR can_borrow()
# ------------------------------------------------
def test_pending_fine_branch():
    assert can_borrow(0, 101, "student") is False
    assert can_borrow(0, 100, "student") is True
def test_faculty_borrowing_branches():
    assert can_borrow(9, 0, "faculty") is True
    assert can_borrow(10, 0, "faculty") is False
def test_non_faculty_borrowing_branches():
    assert can_borrow(2, 0, "student") is True
    assert can_borrow(3, 0, "student") is False
# ------------------------------------------------
# 3. BRANCH TESTS FOR renewal_status()
# ------------------------------------------------
def test_reservation_branch():
    assert renewal_status(0, True, 5) == "denied"
    assert renewal_status(0, False, 5) == "renewed"
def test_renewal_limit_branch():
    assert renewal_status(2, False, 5) == "limit reached"
    assert renewal_status(1, False, 5) == "renewed"
def test_overdue_branch():
    assert renewal_status(1, False, -1) == "overdue"
```

```
    assert renewal_status(1, False, 0) == "renewed"
def test_renewal_condition_priority():
    assert renewal_status(2, True, -1) == "denied"
```